5. Core Actuarial Assumptions & In-Force Forecasting Layer — Humana NPV Model (v4.2.0)

Part 1: Assumption Governance & Register Architecture

Every assumption in the Humana NPV Forecasting Model (v4.2.0) is governed by an 89-entry register (`config/assumption_basis.json` and docs/ASSUMPTION_BASIS.md) and enforced by automated tests:

- **Public Sources (30 entries):** Medicare Trustees, CMS National Health Expenditure (NHE), MACPAC, SSA life tables, NAIC, Milliman, Gen Re, Segal, ADA Health Policy Institute (HPI), state rate guides, Damodaran, NEAM.
- **Statute or Regulation (4 entries):** IRC §11 (21% corporate tax), IRC §848 (9.2% DAC capitalization over 180 months), NAIC Med Supp Model Regulation (65% individual / 75% group lifetime loss ratio standards), 42 CFR 438.8 (85% Federal Medicaid MLR standard).
- **Derived for Internal Consistency (4 entries):** Including Med Supp rate increases solved to match age-curve and trend mechanics.
- **Model Settings (10 entries):** Horizon lengths, cohort sizes, feature switches.
- **Documented Judgment with Sensitivity Ranges (41 entries):** Swept across 32 deterministic sensitivity sweeps (`npv_judgment_sensitivity.csv`), 17 tornado drivers, and 9 Monte Carlo volatility/correlation parameter sweeps (`npv_mc_parameter_sensitivity.csv`).
- **Pricing Policy (v4.2):** The risk tolerance for the pricing solve (P(DE NPV < 0) ≤ 10%) is held in config/pricing_targets.json and governed with the assumption set: the approval hash (AS-SYN-2026-009) covers npv_assumptions.json plus claim_cost_curves.json, trend_benchmarks.json, forecast_config.json and pricing_targets.json, so changing any of them requires re-approval.

Master Comparison of Assumptions Across Lines of Business

Assumption Category	Medicare Supplement (Plan G)	Individual Dental & Vision	Medicaid Managed Care	What Dictates It & How It Is Forecasted
Starting Revenue / Rate	`$1,980/yr` ($165/mo) at age 65; `+3.0%` attained-age step; `+7.3%` annual rate increase	`$468/yr` ($39/mo); annual rate increase tied to trend	Blended Capitation `MD_Cap1` (Child / Adult / ABD cells); `+4.5%` state rate update	**Dictated by:** State DOI rate filings (Med Supp/D&V) and State Actuary rate certifications (Medicaid). In Monte Carlo v4, a **dynamic rate-filing response** adjusts future Med Supp/D&V rate

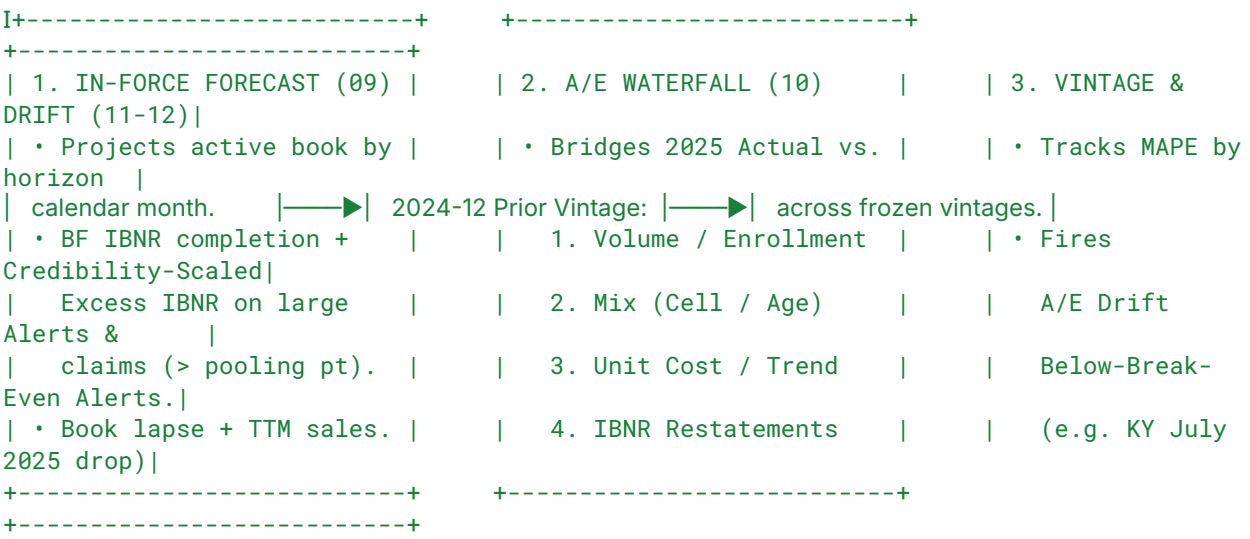
Assumption Category	Medicare Supplement (Plan G)	Individual Dental & Vision	Medicaid Managed Care	What Dictates It & How It Is Forecasted
				filings (with a 2-yr lag) to close emerging trend gaps.
Starting Claim Cost & Morbidity	`MS_Clm1` ultimate cost; `+3.8%` age morbidity slope; Select factors `MS_Sel1..3` (underwriting wear-off)	`DV_Clm1` split across 4 categories (Prev, Basic, Major, Vision); `1.28×` Year-1 Pent-Up factor	`88.0%` Year-1 Medical Cost Ratio (`MD_MCR1`) + Year-1 new-state acuity load (`MD_Acuity`)	**Dictated by:** Chain-ladder completed PMPM experience (`03_completion_factors.sql`, `05_npv_assumption_inputs_gold.csv`) + Bornhuetter-Ferguson (BF) IBNR with credibility-weighted large-claim pooling charges.
Medical / Claim Trend	`6.5%` annual trend (`MS_Trend`)	`3.5%` annual trend (`DV_Trend`)	`4.5%` annual trend (`MD_Trend`)	**Dictated by:** `02_claims_trend_forecast.py`. Fits **normalized** PMPM trends (stripping out aging, duration/selection wear-off, and benefit mix shifts) and reports each indication against the Medicare Trustees / CMS NHE / ADA HPI benchmark. Pricing trend is set from the public benchmark: the synthetic book is not relevant experience (ASOP 25 §3.3), so it gets zero credibility and

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				its indication is a dispositioned REVIEW item. On Humana data the same credibility blend gives company experience its weight.
Lapse, Churn & Persistency	Durational base lapse (`MS_Lapse1..3`) + Anniversary shock (`MS_Shock`) + Dynamic slope (`0.5` per pt of rate inc)	`30%` flat/durational annual lapse (`DV_Lapse1..3`)	`25%` blended annual disenrollment (`MD_Disenroll`) + `2,800/yr` new joiners (`MD_NewMembers`)	**Dictated by:** Poisson GLM lapse study (`03_lapse_glm_study.py`) on exposure months, separating Q1 rate-shock spikes (Med Supp) and removing one-off macro waves like the 2023–2024 Medicaid redetermination unwinding.
Selection & Acuity Dynamics	Shopper share (`MS_ShopShare`), Shopper relative cost (`0.85`, range `0.75–0.95`), Shopper shock propensity	Deferred pent-up demand when waiting periods (`DV_CatW`) are applied	Plan acuity (`MD_Acu`), `80%` Risk-Adjustment capture (`MD_RACap`) with lag (`MD_RALag`), Differential cell churn	**Dictated by:** `Actuarial judgment swept across bounds. At MACPAC relative churn (Child 22.6%, Adult 33.3%, ABD 10.7%), ABD membership doubles from 11.8% to 22.2% of the book, flipping Medicaid DE NPV from -$1.09M to`

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				+$2.71𝕄.
Expenses, Taxes & Capital	FY Comm, Ren Comm, 2% Prem Tax, Acq & Maint costs, 9.2% §848 DAC Tax, Paid-claim RBC %	FY/Ren Comm, Prem Tax, Acq/Maint, 0% DAC Tax, RBC %	2% Quality Withhold (85% earn-back), 85% MLR Floor, Prem Tax, PMPM Admin, Start-up cost, 15% RBC	**Dictated by:** Statutory accounting rules, IRC §11 (21% corporate tax) & §848 (9.2% DAC capitalization over 180 mos), NAIC Med Supp loss ratio standards, 42 CFR 438.8 (85% Medicaid MLR), and 8.5% corporate hurdle rate (DiscRate).

Part 2: The Monthly In-Force Forecasting & Analytics Layer (Steps 09–12)

While Steps 02–06 govern and value new-business cohorts, **Steps 09–12** execute the monthly calendar-year forecasting and monitoring cycle that an **Associate Actuary — Forecasting/Analytics** runs on Humana's existing book of business:



\1. In-Force Calendar-Month Forecast (`09_inforce_forecast.py`)

- **Volume Roll-Forward:** Projects active policies/members using empirical book lapse from the lapse study (rather than pricing lapse assumptions) and projects new sales using trailing 12-month (TTM) sales trend capped at ±30%. (This v4.1 fix eliminated persistent volume bias across all 5 forecast vintages.)
- **Bornhuetter-Ferguson (BF) IBNR & Excess Large-Claim IBNR:**
    - Claims below the pooling threshold are completed using Bornhuetter-Ferguson completion factors (blending chain-ladder paid development with an a-priori PMPM).
    - **Credibility-Weighted Large-Claim Pooling (v4.1 Fix):** In v4.0, a single $250,000 Medicaid claim biased the incurred forecast by +2.8%. Version 4.1 applies **Limited-Fluctuation Credibility** to the empirical pooling charge, where the full-credibility claim count standard is scaled by severity variability:
    - **Credibility-Weighted Large-Claim Pooling (v4.1 Fix):** In v4.0, a single $250,000 Medicaid claim biased the incurred forecast by +2.8%. Version 4.1 applies **Limited-Fluctuation Credibility** to the empirical pooling charge, where the full-credibility claim-count standard is scaled by excess severity variability (CV^2):
        - **Full-Credibility Standard:** $N_{full} = 1{,}082 \times (1 + CV^2_{excess_severity})$
        - **Credibility Weight:** `Z_pool = min(1.0, √(N_observed_large_claims ÷ N_full))`
        - **Blended Pooling Charge:** `Credibility_Pooling_Charge = Z_pool × Empirical_Pooling_Charge + (1 - Z_pool) × Benchmark_Pooling_Charge`
    - **Explicit Excess IBNR Reserve:** `Excess_IBNR = (1 - Completion_Factor) × Credibility_Pooling_Charge × A_Priori_Claims`

2. Actual-vs-Expected (A/E) Variance Waterfall (`10_actual_vs_expected.py`)

Decomposes the dollar variance between realized 2025 calendar-year incurred claims and the prior frozen forecast vintage (`2024-12`) into orthogonal, independently tested steps:

1. **Expected Incurred Claims (Prior Vintage 2024-12)**
2. **+ Enrollment / Volume Variance:** Impact of actual vs. forecasted member months at expected mix and PMPM.
3. **+ Demographic / Rate-Cell Mix Variance:** Shift in attained-age/plan distribution (Med Supp) or Child/Adult/ABD cell mix (Medicaid).
4. **+ Unit Cost / Morbidity / Trend Variance:** Residual pure PMPM claim cost deviation.
5. **+ Prior-Period IBNR Restatement:** Development on prior incurral periods.
6. **= Realized Actual Incurred Claims (Current Vintage)**

3. Vintage Accuracy Tracking & Credibility-Scaled Drift Monitoring (`11_forecast_vintages.py`, `12_drift_monitoring.py`)

- **Vintage Registry:** Freezes each monthly forecast keyed on the `as_of_month`, tracking Mean Absolute Percentage Error (MAPE = 2.3% to 6.7% by horizon) and testing for persistent directional bias across 12 metrics and 5 vintages.
- **Credibility-Scaled A/E Drift Bands:** Raises `REVIEW` or `ALERT` flags when quarterly claims A/E (outside ±5% scaled by exposure credibility) or termination A/E (outside ±15%) persists.
- **End-to-End Data Lineage Example (The Planted Kentucky July 2025 Feed Drop):**
    - In the synthetic dataset, Kentucky Medicaid eligibility is missing for July 2025.
    - Step 01 (`01_data_quality_checks.sql`) flags the missing eligibility month and traces 336 of the 376 orphan claims directly to it.
    - Step 10 (`10_actual_vs_expected.py`) isolates the resulting dollar drop in the A/E Waterfall as a **Volume / Enrollment variance** rather than a false medical morbidity signal.